Beyond the Syllabus — Exploration 02

The Hidden Plane

From Vector Equations to Great Circles

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The question behind the question.

Question 16(c) of the 2026 HSC Mathematics Extension 2 examination concerns circles on a unit sphere. In part (ii), each circle is given by a trigonometric vector equation. A natural approach is to equate the coordinates and solve for two unknown angles. But perhaps the angles are not the most useful quantities to find. Could the geometry tell us which equations to solve instead?

Consider the first circle in the examination:

$$C_1: \quad \mathbf{r}_1(\alpha) = \frac{\cos\alpha}{\sqrt{5}}\begin{pmatrix}1\\2\\0\end{pmatrix} + \sin\alpha\begin{pmatrix}0\\0\\-1\end{pmatrix}, \qquad 0 \le \alpha < 2\pi.$$

Its coordinates are

$$x = \frac{\cos\alpha}{\sqrt{5}}, \qquad y = \frac{2\cos\alpha}{\sqrt{5}}, \qquad z = -\sin\alpha.$$

Before finding any value of α, notice that the first two coordinates always satisfy

$$\boxed{y = 2x.}$$

In three-dimensional space, this is the equation of a *plane*, not a line: z remains unrestricted. Every point on C_1 lies in that fixed plane. We have uncovered a geometric constraint without solving for the parameter.

Why is the path a circle?

The ordinary unit circle is obtained by combining two perpendicular unit directions:

$$\begin{pmatrix}x\\y\end{pmatrix} = \cos\theta\begin{pmatrix}1\\0\end{pmatrix} + \sin\theta\begin{pmatrix}0\\1\end{pmatrix}.$$

The same construction works in three dimensions. If $\mathbf{a}$ and $\mathbf{b}$ are perpendicular unit vectors, write

$$\mathbf{v}(\theta) = \cos\theta\,\mathbf{a} + \sin\theta\,\mathbf{b}, \qquad 0 \le \theta < 2\pi.$$

Then, since $\mathbf{a}\cdot\mathbf{b} = 0$ and $|\mathbf{a}| = |\mathbf{b}| = 1$,

$$\begin{aligned}|\mathbf{v}(\theta)|^2 &= \mathbf{v}(\theta)\cdot\mathbf{v}(\theta)\\ &= \cos^2\theta\,|\mathbf{a}|^2 + 2\sin\theta\cos\theta\,(\mathbf{a}\cdot\mathbf{b}) + \sin^2\theta\,|\mathbf{b}|^2\\ &= \cos^2\theta + \sin^2\theta = 1.\end{aligned}$$

This is the calculation required in part (i). It places the path on the unit sphere, but it does not by itself establish that the path is a circle: the sphere contains many curves. The further observation is that $\mathbf{v}(\theta)$ always lies in the fixed plane spanned by $\mathbf{a}$ and $\mathbf{b}$, and as θ varies it traces the entire unit circle within that plane.

A circle obtained by intersecting a sphere with a plane through its centre is called a *great circle*. Thus, in this setting,

$$\boxed{\text{great circle} = \text{plane through the origin } \cap \text{ unit sphere.}}$$

Two planes, two points.

Two distinct planes through the origin intersect in a line through the origin. That line meets the unit sphere at two opposite, or *antipodal*, points. We can therefore predict that the two distinct great circles in part (ii) have exactly two intersections, $\mathbf{q}$ and $-\mathbf{q}$, before doing any algebra.

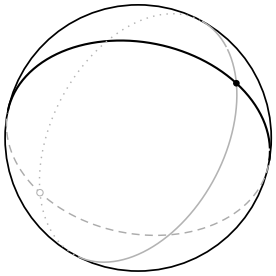


Figure 1. Two great circles intersect at antipodal points. Dashed portions lie on the far side of the sphere.

Returning to Question 16(c)(ii).

The second circle is

$$C_2: \quad \mathbf{r}_2(\theta) = \frac{\cos\theta}{\sqrt{2}}\begin{pmatrix}0\\1\\1\end{pmatrix} + \sin\theta\begin{pmatrix}1\\0\\0\end{pmatrix}, \qquad 0 \le \theta < 2\pi.$$

Here $x = \sin\theta$ and $y = z = \cos\theta/\sqrt{2}$, so every point on C_2 satisfies $z = y$. A common point of the two circles must consequently satisfy

$$\boxed{y = 2x, \qquad z = y, \qquad x^2 + y^2 + z^2 = 1.}$$

The first two equations describe the intersection of the planes; the third confines that line to the unit sphere. Substituting gives

$$x^2 + (2x)^2 + (2x)^2 = 1 \quad \Longrightarrow \quad 9x^2 = 1.$$

Hence the points of intersection are

$$\boxed{\left(\frac{1}{3}, \frac{2}{3}, \frac{2}{3}\right) \quad \text{and} \quad \left(-\frac{1}{3}, -\frac{2}{3}, -\frac{2}{3}\right).}$$

These are antipodal, as predicted. Moreover, the original parametrisations each trace the *whole* unit circle in their containing plane, so every point satisfying the plane and sphere equations belongs to the corresponding curve. No extraneous intersections have been introduced.

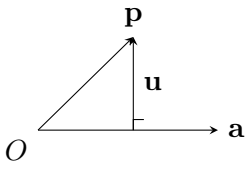
The advantage of recognising the planes is not that we avoid the examination mathematics. Rather, we understand why these simpler simultaneous equations are sufficient.

Constructing a great circle through two points.

Part (iii) reverses the process. Instead of giving two perpendicular unit directions, the question gives two points on the sphere and asks us to construct a circle through them:

$$A = (0, 1, 0), \qquad P = \left(\frac{\sqrt{3}}{2}, \frac{1}{\sqrt{6}}, -\frac{1}{2\sqrt{3}} \right).$$

Take $\mathbf{a} = \overrightarrow{OA} = (0, 1, 0)$ as the first unit direction, and let $\mathbf{p} = \overrightarrow{OP}$. To find another unit direction in the same plane but perpendicular to $\mathbf{a}$, subtract from $\mathbf{p}$ its projection onto $\mathbf{a}$.



Since $|\mathbf{a}| = 1$,

$$\begin{aligned} \mathbf{u} &= \mathbf{p} - (\mathbf{p} \cdot \mathbf{a})\mathbf{a} \\ &= \left(\frac{\sqrt{3}}{2}, \frac{1}{\sqrt{6}}, -\frac{1}{2\sqrt{3}} \right) - \frac{1}{\sqrt{6}}(0, 1, 0) \\ &= \frac{1}{2\sqrt{3}}(3, 0, -1). \end{aligned}$$

Now $|\mathbf{u}| = \sqrt{5/6}$, so the second perpendicular unit vector is

$$\mathbf{b} = \frac{\mathbf{u}}{|\mathbf{u}|} = \frac{1}{\sqrt{10}}(3, 0, -1).$$

The required circle therefore has vector equation

$$\boxed{\mathbf{v}(\theta) = \cos\theta \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + \frac{\sin\theta}{\sqrt{10}} \begin{pmatrix} 3 \\ 0 \\ -1 \end{pmatrix}, \qquad 0 \le \theta < 2\pi.}$$

This contains A when $\theta = 0$, and it contains P for $\cos\theta = 1/\sqrt{6}$ and $\sin\theta = \sqrt{5/6}$. Projection has done more than simplify a calculation: it constructs the perpendicular directions that define the plane of the circle.

One limit is revealing: if $\mathbf{p} = -\mathbf{a}$, the perpendicular component is $\mathbf{u} = \mathbf{0}$ and cannot be normalised. This reflects the geometry: two antipodal points lie on infinitely many great circles and cannot determine a unique plane.

Taking the investigation into a classroom.

After introducing three-dimensional vector equations, show C_1 and ask students what remains true of its coordinates as α varies. Reveal the plane $y = 2x$ with the interactive, then introduce C_2 and ask students to predict the intersections before solving part (ii). For part (iii), ask how projection creates a perpendicular direction without leaving the plane of the given points. The central teaching point is to recognise the geometry *before* choosing the algebra.

Explore the interactive.

Rotate the sphere, move a point around the first circle, and reveal the two planes and their intersections. **Open the interactive online**, or scan the QR code.

Source. Inspired by Question 16(c), 2026 HSC Mathematics Extension 2 examination, p. 19, NSW Education Standards Authority (NESA). Official examination archive. The geometric investigation, diagrams and explanations are independently developed.